

Contact

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Top Skills

Terraform

Docker

OpenShift

Ramaakshay Mallireddy

SWE @ IBM | ASE @ UT Austin | Class of 2029

Austin, Texas, United States

Experience

IBM

Software/AI Engineering Intern

May 2026 - Present (3 months)

Austin, TX

Designed AI-powered hybrid cloud solutions for enterprise clients as part of IBM's Client Engineering team

Implemented cloud-native workloads using Python/Java, containerization (Docker/Kubernetes), and orchestration on Red Hat OpenShift and IBM Cloud platforms

Collaborated with cross-functional teams to translate client requirements into scalable technical architectures and proof-of-concept deliverables

The University of Texas at Austin

Texas Spacecraft Laboratory SCOPE-1 Electrical Power Systems Member

August 2025 - Present (1 year)

Austin, Texas, United States

- Selected and tested over 15+ power components, including batteries, solar panels, and converters, ensuring compatibility with spacecraft power requirements.
- Developed and ran orbital energy simulations predicting power consumption within $\pm 5\%$ accuracy across mission scenarios.
- Designed and validated 3+ custom PCBs supporting subsystem power distribution and DC-DC conversion, improving system efficiency by $\sim 10\%$.

Aerospace Engineering and Engineering Mechanics, The University of Texas at Austin

AIAA Aircraft Design Team Propulsion Engineer

August 2025 - Present (1 year)

Selected and integrated 2 high-thrust jet engines to achieve >250 N total thrust, enabling catapult-assisted launch and sustained Mach 1.6 dash performance for the DC25 aircraft.

Modeled and optimized fuel and thermal management systems using MATLAB and ANSYS, reducing avionics operating temperature by 18% and increasing system endurance by 22%.
Coordinated with aerodynamics and structures teams to refine engine nacelle geometry and intake placement, improving thrust-to-weight ratio by 15% and cutting drag by 9%

Austin Energy

Engineer Intern

June 2024 - August 2024 (3 months)

Austin, Texas, United States

Played a pivotal role in strategizing the layout of urban energy infrastructure, orchestrating meticulous blueprints for city energy grids. Executed numerous on-site implementations, translating conceptual plans into tangible realities, thereby fortifying the urban landscape with robust energy systems. Implemented grid power automation for transmission lines and transformers. Ran many test cases in AUD to properly upgrade the current AUD version being used. Utilized AUD to plan out and execute a design plan of a complex portion of the Austin electrical grid.

The Aerospace Corporation

Aerospace Engineer Intern

June 2024 - August 2024 (3 months)

Austin, Texas, United States

Athena EdTech

Student Ambassador

June 2024 - August 2024 (3 months)

Honored to be chosen among hundreds of applicants with a selective acceptance rate of 10%. As a student ambassador GPT Developer at Athena EdTech's non-profit branch, I advocate for the integration of accessible AI technologies in K-12 classrooms and educate educators on their use. My role involves supporting the adoption of AI in education through advocacy and partnership initiatives. Additionally, I develop various GPT models to enhance classroom experiences with AI integration.

Elysium Robotics

Mechanical Engineering Intern

May 2024 - July 2024 (3 months)

Austin, Texas, United States

Developed simulations for robotic hand motion, designed and produced advanced robotic muscles, and collaborated with engineers to integrate them into a humanoid robot.

Self-employed

Scientific Researcher

September 2023 - May 2024 (9 months)

Austin, Texas, United States

Completed and disseminated a comprehensive research endeavor elucidating methodologies for detecting celestial bodies via sophisticated statistical analyses and rigorous experimentation.

Self-employed

Software Programmer

May 2023 - June 2023 (2 months)

Austin, Texas, United States

Developed a planetary simulation implementing real time data and astrophysics | Implementing this simulation in several RRISD Astronomy courses

Education

Mcneil High School

· (2021 - 2025)

The University of Texas at Austin

Bachelor of Engineering, Aerospace, Aeronautical and Astronautical Engineering